# noc22_cs107

**Mock Exam 3 | Course ID: noc22_cs107**

### Q1. According to the lecture, the "software crisis" refers to a persistent set of problems in software development. Which of the following is NOT a symptom of this crisis?

A) Projects exceeding their budget.
B) Software that is difficult to alter or maintain.
C) Hardware costs increasing faster than software costs.
D) Projects being delivered late.

*Source: ilHPCbkZLV4 | Difficulty: Easy*

### Q2. What is the primary purpose of conducting a feasibility study at the beginning of a project?

A) To write the initial lines of code for the project.
B) To determine if the project is viable and should be undertaken.
C) To hire the entire development team.
D) To finalize the user interface design.

*Source: yC0A6_nYPYQ | Difficulty: Easy*

### Q3. The "Build and Fix Model" is described as being suitable for which type of projects?

A) Large, complex projects with many team members.
B) All projects that require high reliability.
C) Trivial projects, such as student assignments.
D) Projects with a fixed, unchangeable budget.

*Source: A8-PtgugnGE | Difficulty: Easy*

### Q4. In the Scrum framework, what is a "Sprint"?

A) A daily 15-minute meeting for the development team.
B) The final phase of the project where all testing is done.
C) A prioritized list of all desired features for a product.
D) A fixed-duration iteration during which a product increment is developed.

*Source: 0JYkgfgKn8I | Difficulty: Easy*

### Q5. What does the acronym PERT stand for in the context of project scheduling?

A) Project Execution and Resource Tracking.
B) Program Evaluation and Review Technique.
C) Product Engineering and Revision Tool.
D) Process Enhancement and Reporting Technology.

*Source: B6Awvb58gIA | Difficulty: Easy*

### Q6. What is the primary goal of the "Price to Win" estimation strategy?

A) To accurately calculate the total effort required for the project.
B) To set the project cost based on what the customer is willing to spend.
C) To estimate the project size using Function Points.
D) To ensure the highest possible quality for the final product.

*Source: wqyJF-Gbdp0 | Difficulty: Easy*

### Q7. According to the lecture, what is a primary use of a Gantt chart in project management?

A) To write and compile source code.

B) To conduct user acceptance testing.
C) To plan the utilization of various resources.
D) To formally close a contract with a vendor.
*Source: bvt6NAabXPA | Difficulty: Easy*

### Q8. What is the core concept of "egoless programming" within a democratic team structure?

A) Only the most senior programmer is allowed to write code.
B) No single individual "owns" a piece of code; work is reviewed by all.
C) Programmers work in complete isolation to avoid conflicts.
D) All technical decisions are made by the project manager.
*Source: MpNjo7bhBTg | Difficulty: Easy*

### Q9. What is the fundamental difference between a "fault" and a "failure" in software testing?

A) A fault is a human mistake, while a failure is a bug in the code.
B) A fault is an incorrect step in the code, while a failure is the observable deviation from expected behavior.
C) A fault can only be found by the developer, while a failure is found by the user.
D) A fault and a failure are synonyms for the same problem.
*Source: Qfy0niH--tk | Difficulty: Easy*

### Q10. What does ISO 9001 certification primarily signify about an organization?

A) That its individual software products are guaranteed to be bug-free.
B) That the organization has a documented quality process or system.
C) That the organization is the most profitable in its industry.
D) That all its employees have passed a specific quality exam.
*Source: ODQNex8xtjU | Difficulty: Easy*

### Q11. According to Barry Boehm's W5HH principle for defining project scope, which question specifically addresses the project's budget and resource requirements?

A) Why is the software being built?
B) What will be done?
C) Who is responsible for a function?
D) How much of each resource is needed?
*Source: nmkKAZ0jatM | Difficulty: Medium*

### Q12. In the V-Model of software development, the design of integration test cases is typically carried out in parallel with which development phase?

A) Requirements Specification
B) High-Level Design
C) Detailed Design
D) Coding
*Source: J1-LpZr9GSQ | Difficulty: Medium*

### Q13. What is the key difference between a purely incremental model and an evolutionary model?

A) The incremental model uses iterations, while the evolutionary model does not.
B) The evolutionary model delivers working software to the customer, while the incremental model does not.
C) In the incremental model all requirements are defined upfront, while in the evolutionary model they emerge over time.
D) The incremental model is for large projects, while the evolutionary model is only for small projects.
*Source: sHoPCWz6YeQ | Difficulty: Medium*

**Q14. Which project evaluation technique calculates the difference between total income and total costs but is criticized for not accounting for the timing of cash flows?**

A) Net Present Value (NPV)
B) Internal Rate of Return (IRR)
C) Payback Period
D) Net Profit

*Source: 8UlgpszPK6w | Difficulty: Medium*

**Q15. What is a major shortcoming of using Source Lines of Code (SLOC) as a project size estimation metric?**

A) It is too difficult for non-technical stakeholders to understand.
B) It can only be used for projects written in assembly language.
C) It penalizes the use of high-level languages and code reuse.
D) It requires expensive, specialized tools to calculate.

*Source: wqyJF-Gbdp0 | Difficulty: Medium*

**Q16. In the COCOMO 81 model, how does the "Embedded Mode" of development typically differ from the "Organic Mode"?**

A) Embedded mode projects are smaller and use experienced teams, while organic mode projects are large and innovative.
B) Embedded mode is for projects with flexible constraints, while organic mode has tight constraints.
C) Embedded mode projects are completely new and innovative, while organic mode projects are familiar and stable.
D) The constants used for effort estimation are lower for embedded mode than for organic mode.

*Source: sQyR78dlSzc | Difficulty: Medium*

**Q17. What is the primary goal of "resource smoothing" in project management?**

A) To increase the total number of resources available to the project.
B) To ensure all activities start on their earliest possible date.
C) To create a more even and consistent demand for resources over time.
D) To identify the project's critical path.

*Source: ZcO5UNzMtz8 | Difficulty: Medium*

**Q18. In PERT/CPM, when calculating the Earliest Start (ES) time for an activity that has multiple predecessors (a merge point), what rule is applied?**

A) The ES is the average of the Earliest Finish (EF) times of all its predecessors.
B) The ES is the earliest (minimum) of the EF times of all its predecessors.
C) The ES is the latest (maximum) of the EF times of all its predecessors.
D) The ES is always set to zero.

*Source: 8w2gm0AO-KY | Difficulty: Medium*

**Q19. According to Herzberg's Two-Factor Theory, what is the primary role of "hygiene factors" such as pay and company policy in employee motivation?**

A) They are the main drivers of job satisfaction and high performance.
B) They have no impact on employee satisfaction or dissatisfaction.
C) Their absence causes dissatisfaction, but their presence does not actively motivate.
D) They are only important for employees at the self-actualization level.

*Source: TQWJYqOzs0g | Difficulty: Medium*

**Q20. A project has an Earned Value (EV) of $50,000, a Planned Value (PV) of $60,000, and an Actual Cost (AC) of $55,000. Based on these figures, what is the status of the project?**

A) Ahead of schedule and under budget.
B) Behind schedule and over budget.

C) Ahead of schedule and over budget.
D) Behind schedule and under budget.
*Source: I81fOjyzb0E | Difficulty: Medium*

**Q21. What is the primary advantage of using a restricted tendering process compared to an open tendering process for contractor selection?**

A) It guarantees the lowest possible price due to maximum competition.
B) It is mandated by law for all government projects.
C) It reduces the evaluation time and effort by soliciting bids only from pre-selected suppliers.
D) It eliminates the need for a formal contract.
*Source: olrW1tqGnsM | Difficulty: Medium*

**Q22. In the SEI CMM framework, what is the key characteristic that distinguishes a Level 3 (Defined) organization from a Level 2 (Repeatable) organization?**

A) Level 3 uses quantitative quality goals, while Level 2 does not.
B) Level 3 has processes that are documented and standardized across the organization, while Level 2's processes are project-specific.
C) Level 3 relies on individual heroics, while Level 2 has basic project management.
D) Level 3 focuses on continuous process improvement, while Level 2 focuses on repeating success.
*Source: kZ0HN-9zdik | Difficulty: Medium*

**Q23. A project manager uses the Value-to-Cost (V/C) ratio to prioritize increments. Given the following functionalities and their customer-rated value (V) and developer-rated cost (C), which should be developed first? Profit Reports (V=9, C=2), Online Database (V=1, C=10), Ad Hoc Enquiry (V=5, C=5), Profit-Based Pay (V=9, C=1).**

A) Profit Reports
B) Online Database
C) Ad Hoc Enquiry
D) Profit-Based Pay
*Source: 0O6fJSkafOo | Difficulty: Hard*

**Q24. In the context of the Delphi estimation technique, what is the primary purpose of ensuring anonymity among experts during the estimation rounds?**

A) To save time by avoiding lengthy group discussions.
B) To prevent a dominant personality from unduly influencing the group's estimates.
C) To make it easier for the coordinator to calculate the final average.
D) To allow experts to use different estimation formulas without conflict.
*Source: wqyJF-Gbdp0 | Difficulty: Hard*

**Q25. A project is estimated using the Albrecht/IFPUG method. It has 50 user inputs, 40 user outputs, 35 inquiries, 6 internal files, and 4 external interfaces, all of average complexity. If the Total Degree of Influence (TDI) is calculated to be 42, what is the final Adjusted Function Point (AFP) count? (Use average weights: EI=4, EO=5, EQ=4, LIF=10, EIF=7).**

A) 628
B) 672
C) 618
D) 661
*Source: rOn-GA71uKM | Difficulty: Hard*

**Q26. According to Putnam's estimation model, the relationship between effort (K) and development time (td) is K = C1 / td^4. If a project's schedule is compressed to half of its original planned time, by what factor will the total effort and cost increase?**

A) 2 times
B) 4 times

C) 8 times
D) 16 times
*Source: _KdYrzzMuBo | Difficulty: Hard*

**Q27. A project manager needs to crash a project. The current critical path is A-B-D with a duration of 25 weeks. A parallel path, A-C-E, has a duration of 24 weeks. The costs to crash each task per week are: A=$100, B=$150, C=$80, D=$120. Which task should the manager crash first to reduce the project duration by one week?**

A) Task A, because it is the cheapest on the critical path.
B) Task C, because it is the cheapest overall.
C) Task D, because it is on the critical path.
D) Tasks A and C must be crashed simultaneously.
*Source: ikOTZBPaklE | Difficulty: Hard*

**Q28. What is the primary reason that the hardware "bathtub curve" of reliability, which includes burn-in, useful life, and wear-out phases, is not directly applicable to software?**

A) Software is more complex than hardware.
B) Software failures are caused by design flaws (bugs), not physical wear and tear.
C) Software reliability cannot be measured, whereas hardware reliability can.
D) Hardware warranties cover the burn-in period, but software has no warranties.
*Source: 5iuEGNEyAPU | Difficulty: Hard*

**Q29. During a formal code inspection, which of the following error types is a primary target for discovery, as opposed to an error that is typically caught by a compiler?**

A) A missing semicolon at the end of a statement.
B) Use of an uninitialized variable in a calculation.
C) An incorrectly spelled keyword like "whle" instead of "while".
D) A function call with the wrong number of arguments.
*Source: 4ZLbpoBcdo8 | Difficulty: Hard*

**Q30. What is the core assumption of Quality Assurance (QA) that represents a fundamental shift from earlier Quality Control (QC) methods?**

A) That testing every single product is the only way to ensure quality.
B) That a good, well-defined process, if followed rigorously, will inherently lead to quality products.
C) That statistical methods are not useful for monitoring product quality.
D) That defects should be identified and eliminated only after the product is fully manufactured.
*Source: zQXss82WudM | Difficulty: Hard*

# Answer Key

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|---|---|---|---|---|
| 1: C | 2: B | 3: C | 4: D | 5: B |
| 6: B | 7: C | 8: B | 9: B | 10: B |
| 11: D | 12: B | 13: C | 14: D | 15: C |
| 16: C | 17: C | 18: C | 19: C | 20: B |
| 21: C | 22: B | 23: D | 24: B | 25: B |
| 26: D | 27: A | 28: B | 29: B | 30: B |